Deep Learning for NLP II

Basics
Introduction and Course Outline

Learning goals

- Understand the scope of the course
- Answers to all open question
- Get an impression of the workload

PEOPLE

Lecturers and Tutors:

- Prof. Dr. Dr. Valentin Hofmann
- Dr. Matthias Aßenmacher
- Lea Hirlimann, M.Sc. (exercise sessions)

Time:

- Lecture: Wednesday, 10.00 - 12.00
- Exercise: Friday, 10.00 - 12.00

STRUCTURE OF THE LECTURE

Central building blocks of the lecture

- Transformer Recap (1 week)
- Tokenization (1 week)
- Architecture Variants (3 weeks)
- Advanced LLM Inference (1 week)
- Pretraining, Finetuning, Posttraining (1 week each)
- Reasoning Models (1 week)
- Evaluation and Benchmarking (1 week)
- Alignment, Bias, and Safety (1 week)
- Mechanistic Interpretability (1 week)

STRUCTURE OF THE EXERCISE

Coding Assignments:

- Tokenization
- Retrieval-Augmented Generation
- Speculative Decoding
- LoRa Finetuning
- SFT vs. DPO
- Evaluation and Benchmarking

STRUCTURE OF THE EXERCISE

Reading Assignments:

- ▸ Illustrated Transformer
- ▸ Lost in the Middle
- ▸ H-Net
- ▸ RegMix
- ▸ LIMA
- ▸ ROME

PREREQUISITES

Most important: **Deep Learning for NLP I**

Machine learning basics:

- A proper understanding of
  - linear algebra
  - loss functions
  - regularization
  - classification vs. regression
  - backpropagation and gradient descent
  - simple neural networks (MLPs)
- Great resource: *I2ML course*

MANAGING EXPECTIONS

What to expect

- Insights into advanced concepts and models of contemporary NLP
- Challenging exercises that deepen your understanding, i.e.
  - you will have to invest quite some time in trying to solve them
  - just checking our solutions won't get you far
- Critical discussions of the relevant research papers
- An (inter)actively taught course with motivated lecturers, i.e.
  - we expect you to actively participate in the lecture and ask questions
  - you won't get much out of this course by just enrolling and not showing up

MANAGING EXPECTIONS

What not to expect

- An agentic loop engineering course
- Riding the hypetrain
- Only the latest state-of-the-art LLMs (you also need to know the relevant basics to understand what is going on)
- Low workload and easy exercises (you will only learn this thoroughly by investing some time)

MANAGING EXPECTIONS

Honesty

- No intention to overload you with (unnecessary) work
- Building up proper skills requires **time investment**
- You won't get much out of the coding assignments if you just paste them into your coding agent
- Up-front communication of what is going to be required

MANAGING EXPECTIONS

Workload

- This is a 6 ECTS course
- 1 ECTS := 30h (i.e. this course = 180h)
- 13 weeks $\times$ 4h presence $\approx$ 52h
- 130h for preparation, self-study, **working on assignments**, and **actually reading papers**
- 130h / 14 weeks = **9h** / **week** (*additionally to in-class time*)

Any Questions?